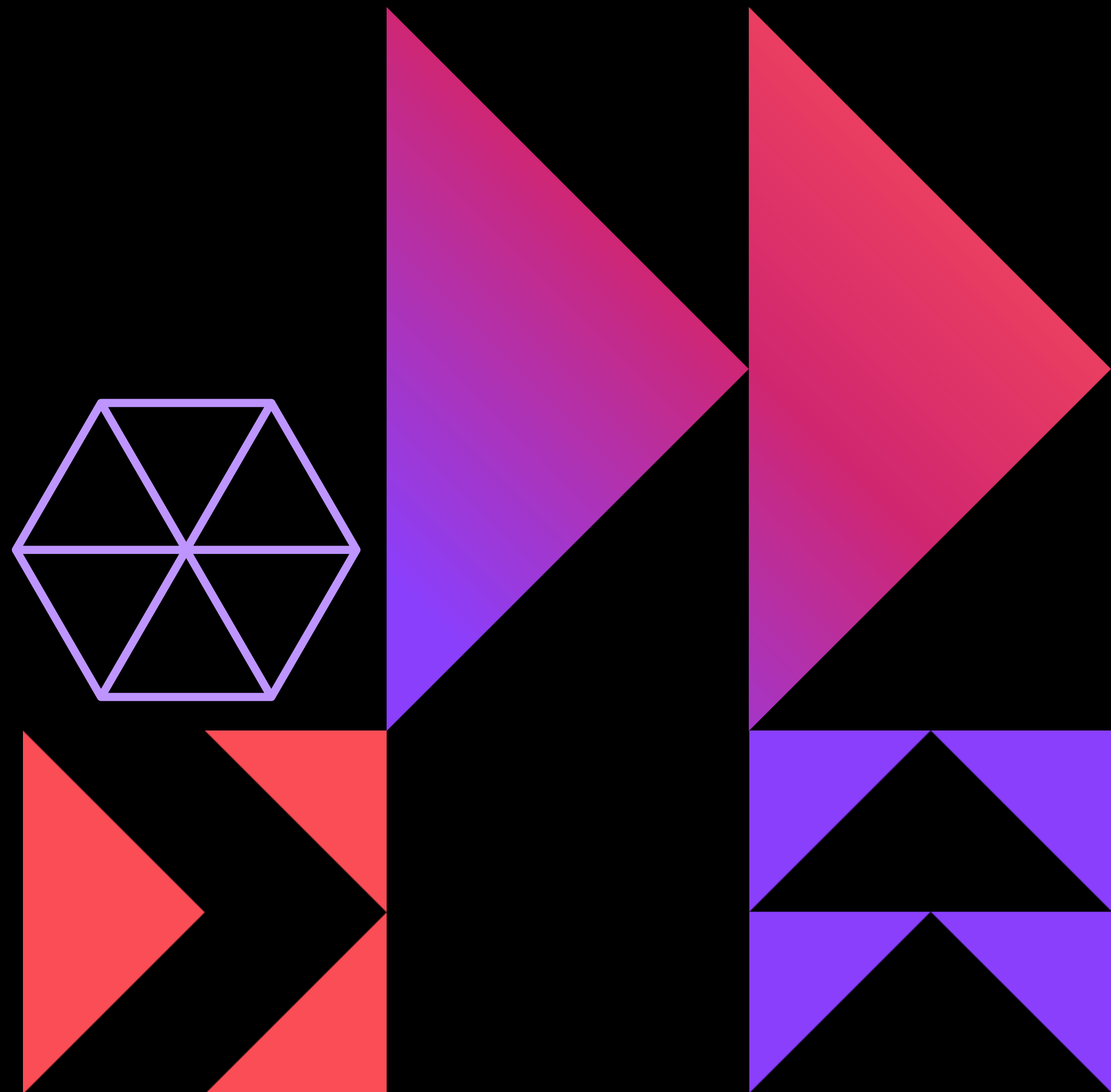
Instructor-led
Lab session (2490): Introduction

Unified Observability and Autonomous Optimization: Hands-on with IBM Instana, Turbonomic, and Cloudability



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The lab journey

From application telemetry to business value

Unified Observability & Autonomous Optimization

Hands-on with IBM Instana, Turbonomic, Cloudability & Apptio AI-TCO

TECHXCHANGE 2026
Session ID: LAB-2490

GOAL: Connect real-time observability to autonomous optimization and financial outcomes.
Students deploy a microservices app, observe it, optimize it, analyze cost and generate an AI-driven TCO business case.

IBM BOB ID (SSO) — Single Sign-On

One identity for TechZone, Jump Host & IBM SaaS Services

Simplified Access Everywhere

Secure & Isolated Experience

STUDENT JOURNEY (WHAT ATTENDEES DO)

1. **Deploy** — Deploy Robot Shop on Red Hat OpenShift
2. **Observe** — IBM Instana auto-discovers services, traces, metrics and infrastructure
3. **Optimize** — IBM Turbonomic analyzes telemetry and recommends / executes rightsizing actions
4. **Analyze Cost** — IBM Cloudability shows cloud spend and rightsizing impact
5. **Generate AI-TCO** — IBM Apptio AI-TCO builds TCO scenarios and generates business case (savings & ROI)

ENVIRONMENT ARCHITECTURE (HOW IT IS BUILT)

PRECONFIGURED LAB WORKSPACE (JUMP HOST / VSI)

- oc CLI, kubectl, Helm
- Preconfigured oc login & kubeconfig
- Sample YAML Manifests
- Helm Charts & Values
- Helper Scripts (deploy.sh, cleanup.sh)
- Browser Bookmarks to All SaaS Portals
- Local Git Repository (Lab Code & Docs)
- Lab Guide

SHARED OPENSHIFT CLUSTER
Large (184 vCPU) – IBM Cloud (us-south)
ISOLATED STUDENT NAMESPACES

STUDENT NAMESPACE → Robot Shop Microservices

- Optimization Signals → **IBM Instana (APM)** — Traces, Metrics, Infrastructure Correlation
- Cost & Utilization Data → **IBM Turbonomic (Autonomous Optimization)** — Recommendations & Automated Rightsizing
- Modeling Data → **IBM Cloudability (FinOps)** — Cloud Spend, Waste, Allocation & Rightsizing Impact
- **IBM Apptio AI-TCO** — AI-driven TCO Scenarios & Business Case (ROI)

SHARED SAAS SERVICES (ACROSS ALL STUDENTS)

- IBM Instana SaaS (1 Tenant)
- IBM Turbonomic SaaS (Essentials)
- IBM Cloudability SaaS (FinOps)
- IBM Apptio AI-TCO SaaS (Managed)

ENVIRONMENT SUMMARY

PER STUDENT

- IBM Bob ID (Single Sign-On)
- Jump Host (Preconfigured)
- Isolated OpenShift Namespace
- Instana Agent Key
- Turbonomic Namespace Scope
- Cloudability Business Unit View
- Apptio AI-TCO Model
- Lab Guide

SHARED (ALL STUDENTS)

- Red Hat OpenShift Cluster (Large)
- IBM Instana SaaS
- IBM Turbonomic SaaS
- IBM Cloudability SaaS
- IBM Apptio AI-TCO SaaS

KEY FACTS

- **DURATION** — 90 MINUTES
- **INFRASTRUCTURE** — SHARED OCP LARGE CLUSTER
- **ISOLATION** — NAMESPACE-LEVEL RBAC & QUOTAS
- **AUTONOMATION** — AUTONOMOUS OPTIMIZATION
- **FINOPS + AI** — COST VISIBILITY & AI-DRIVEN TCO

TAKEAWAY

OBSERVE → OPTIMIZE → QUANTIFY → BUSINESS VALUE & ROI



The lab journey

From application telemetry to business value

1. DEPLOY
   - Deploy Robot Shop microservices on Red Hat OpenShift.

2. OBSERVE
   - IBM Instana automatically discovers services, collects traces and metrics, and correlates application and infrastructure data.

3. OPTIMIZE
   - IBM Turbonomic analyzes resource utilization and provides autonomous rightsizing recommendations and actions.

4. QUANTIFY
   - IBM Cloudability connects optimization decisions to cloud spend, waste, allocation, and governance.



Instana

Move from complexity to confidence with automation

Challenges from IT complexity	Solution: IBM Instana Observability
Siloed data sets, stack complexity	Real-time, full-stack observability
Manual configuration and maintenance	Automated discovery with contextual insights
Non-deterministic systems	Support for modern, cloud-native architectures & (Gen)AI apps
4+ hour mean time to resolution (MTTR)	Reduce incident response time by 65%[1]
Limited automation, incomplete context	Automated instrumentation and dynamic discovery
Operational friction	Actionable insights for strategic, business-level decisions

IBM Turbonomic

247% ROI and <6 months payback

Hybrid Cloud Optimization

Manage, automate and continuously optimize your hybrid cloud



1 Understand Performance Needs

2 Identify Waste

3 Automate Optimization

Hybrid cloud

Multicloud

PaaS/containers

- Get 30% of your engineering time back.
- Assure application performance.
- Improve cloud investments by 35%
- Reduce data center refresh costs by 75%
- Use less carbon by spending less on IT

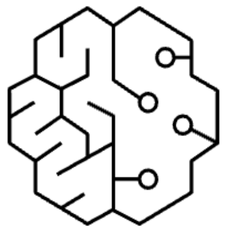
IBM Turbonomic Key Features

Full-stack visualization



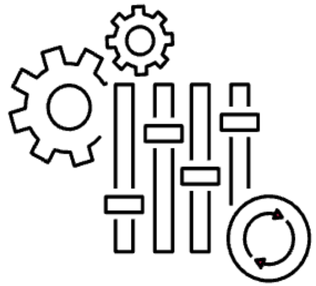
Complete visualization of your applications and infrastructure to stop guessing and start knowing the health of your environment

Analytical insights



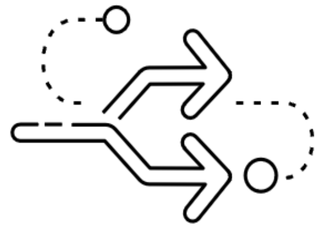
Patented and predictive analytics provide insights that empower teams to make informed decisions to enhance efficiency without sacrificing performance

Operationalize elasticity



Operationalize elasticity to size up or down on demand to assure application performance at the lowest cost

Comprehensive integrations



Seamlessly integrate with your organizations existing tools, enabling rapid time to value

Apptio unblocks the FinOps blockers

Inconsistent tagging and costing across various cloud providers	**View and allocate costs** across a multi-cloud environment
Unable to easily detect cloud waste, anomalies, orphaned / idle resources	**Reduce waste** caused by overprovisioned and idle cloud resources
Cannot easily compare multi-cloud discount programs	**Maximize and automate** coverage and utilization of commitments
Can't assign or reduce container costs in the cloud	Manage and optimize the cost of **containerized infrastructure**
Can't determine **complete set of cost drivers** or **allocate shared costs accurately**	Calculate the **total cost of running products in the cloud** and allocate to teams consuming
Unable to **connect cost and revenue data** holistically	Measure unit cost and **mature to unit economics**

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Thank you

